

A cluster of amyotrophic lateral sclerosis in New Hampshire: a possible role for toxic cyanobacteria blooms.



Cyanobacteria produce many neurotoxins including beta-methylamino-L-alanine (BMAA) that has been linked to amyotrophic lateral sclerosis (ALS) and neurodegenerative disease. A number of ALS cases have been diagnosed among residents of Enfield, NH, a town encompassing a lake with a history of cyanobacteria algal blooms. To investigate an association between toxic cyanobacterial blooms in New Hampshire and development of ALS, we reviewed records from our institution and other community databases to obtain demographic information on patients diagnosed with ALS within New England. We identified nine ALS patients who lived near Lake Mascoma in Enfield, NH, an incidence of sporadic ALS that is 10 to 25 times the expected incidence of 2/100,000/year. We suggest that the high incidence of ALS in this potential cluster could be directly related to chronic exposure to cyanobacterial neurotoxins such as BMAA. Possible routes of toxin exposure include inhalation of aerosolized toxins, consuming fish, or ingestion of lake water. Further investigation, including analysis of brain tissue for cyanobacterial toxins, will be helpful to test for an association between BMAA and ALS.